

Assignment 2

Total: 25

This assignment must be handwritten. Show ALL steps in ALL questions.

1. [5 Marks] **Find** the complement of the following expression:
$$(x' + y + z')(x' + y')(x + z')$$
2. [5 + 5 Marks] **Find** out the Canonical SOP and POS for the following: (Do not use truth table)
 - a. $F(A,B,C,D) = A + B'CD'$
 - b. $F(A,B,C,D,E) = AB + CDE$
3. [5 Marks] **Simplify** the boolean equation $F(a,b,c,d) = \sum(8, 9, 10, 11, 13, 15)$ using the laws of boolean algebra and implement the simplified function using only NOR gates.
4. [5 Marks] **Simplify** the boolean equation $F(a,b,c,d) = \sum(5, 8, 9, 12, 15)$ using the laws of boolean algebra and implement the simplified function using only NAND gates.